



DEPARTAMENTO DE
MATEMÁTICA

The Principles of Deepn Learning Theory

Capítulo 4: RG-flow de pré-ativações

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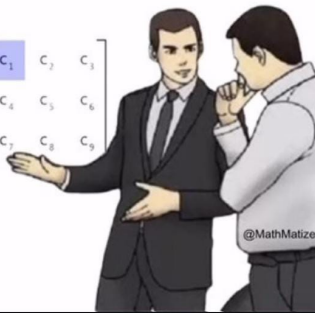
Wick Wick Wick: derivação combinatória

Schwinger-Dyson this way: derivação algébrica

slaps roof of matrix multiplication

This bad boy is behind all of AI

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$



Dataset com inputs n_0 -dimensionais e com N_D amostras:

$$D = \{x_{i,\alpha}\}_{i=1,\dots,n_0;\alpha=1,\dots,N_D}$$

Pré-ativações da 1ª camada são dadas por:

$$z_{i,\alpha}^{(1)} \equiv z_i^{(1)}(x_\alpha) = b_i^{(1)} + \sum_{j=1}^{n_0} W_{ij}^{(1)} x_{j,\alpha}, \quad \text{para } i = 1, \dots, n_1.$$

Na inicialização, estabelecemos $b^{(1)}, W^{(1)}$ independentemente distribuídas e tais que:

$$\mathbb{E}[b_i^{(1)}] = \mathbb{E}[W_{i_1 j_1}^{(1)}] = 0$$

e

$$\mathbb{E}[b_i^{(1)} b_j^{(1)}] = \delta_{ij} C_b^{(1)}$$

$$\mathbb{E}[W_{i_1 j_1}^{(1)} W_{i_2 j_2}^{(1)}] = \delta_{i_1 i_2} \delta_{j_1 j_2} \frac{C_W^{(1)}}{n_0}$$

Estamos interessados em:

$$p(z^{(1)}|D) = p(z^{(1)}(x_1), \dots, z^{(1)}(x_{N_D}))$$

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Vamos fazer a demonstração via 2 caminhos:

- ▶ combinatório, via contrações de Wick e correlatores;
- ▶ algébrico, via transformada de Hubbard-Stratonovich.

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Vemos primeiramente que:

$$\begin{aligned}\mathbb{E}[z_{i,\alpha}^{(1)}] &= \mathbb{E}\left[b_i^{(1)} + \sum_{j=1}^{n_0} W_{ij}^{(1)} x_{j,\alpha_1}\right] \\ &= \mathbb{E}[b_i^{(1)}] + \sum_{j=1}^{n_0} \mathbb{E}[W_{ij}] x_{j,\alpha} = 0\end{aligned}$$

dado que as distribuições tem média 0.

É “fácil ver que” todos os correlatores de ordem ímpar de $p(z^{(1)}|D)$ são zerados, porque sempre sobra um $b^{(1)}$ ou $W^{(1)}$ em cada termo que zera a conta.

Para um correlator de dois pontos:

$$\mathbb{E} \left[z_{i_1; \alpha_1}^{(1)} z_{i_2; \alpha_2}^{(1)} \right] = \mathbb{E} \left[\left(b_{i_1}^{(1)} + \sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(b_{i_2}^{(1)} + \sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right]$$

Para um correlator de dois pontos:

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 \mathbb{E} \left[z_{i_1; \alpha_1}^{(1)} z_{i_2; \alpha_2}^{(1)} \right] &= \mathbb{E} \left[\left(b_{i_1}^{(1)} + \sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(b_{i_2}^{(1)} + \sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right] \\
 &= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1, j_1}] x_{j_1, \alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2, j_2}] x_{j_2, \alpha_2} \\
 &\quad + \mathbb{E} \left[\left(\sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(\sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right]
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 &= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1, j_1}] x_{j_1, \alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2, j_2}] x_{j_2, \alpha_2} \\
 &\quad + \mathbb{E} \left[\left(\sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(\sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right] \\
 &= \delta_{i_1 i_2} C_b^{(1)} + \sum_{j_1, j_2=1}^{n_0} \mathbb{E}[W_{i_1 j_1} W_{i_2 j_2}] x_{j_1, \alpha_1} x_{j_2, \alpha_2}
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 &= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1, j_1}] x_{j_1, \alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2, j_2}] x_{j_2, \alpha_2} \\
 &\quad + \mathbb{E} \left[\left(\sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(\sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right] \\
 &= \delta_{i_1 i_2} C_b^{(1)} + \sum_{j_1, j_2=1}^{n_0} \mathbb{E}[W_{i_1 j_1} W_{i_2 j_2}] x_{j, \alpha_1} x_{j, \alpha_2} \\
 &= \delta_{i_1 i_2} \left(C_b^{(1)} + C_W^{(1)} \frac{1}{n_0} \sum_{j=1}^{n_0} x_{j; \alpha_1} x_{j; \alpha_2} \right)
 \end{aligned}$$

Definimos a seguinte expressão como a **métrica de primeira camada**:

$$G_{\alpha_1\alpha_2}^{(1)} := C_b^{(1)} + C_W^{(1)} \frac{1}{n_0} \sum_{j=1}^{n_0} x_{j;\alpha_1} x_{j;\alpha_2}$$

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Note que é uma função de $x_{\alpha_1}, x_{\alpha_2}$. Assim, nossa igualdade acima se traduz como:

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Note que é uma função de $x_{\alpha_1}, x_{\alpha_2}$. Assim, nossa igualdade acima se traduz como:

$$\mathbb{E} \left[z_{i_1;\alpha_1}^{(1)} z_{i_2;\alpha_2}^{(1)} \right] = \delta_{i_1 i_2} G_{\alpha_1\alpha_2}^{(1)}$$

Seguindo a mesma ideia:

$$\begin{aligned}\mathbb{E}[z_{i_1;\alpha_1}^{(1)} z_{i_2;\alpha_2}^{(1)} z_{i_3;\alpha_3}^{(1)} z_{i_4;\alpha_4}^{(1)}] &= \delta_{i_1 i_2} \delta_{i_3 i_4} G_{\alpha_1 \alpha_2}^{(1)} G_{\alpha_3 \alpha_4}^{(1)} + \delta_{i_1 i_3} \delta_{i_2 i_4} G_{\alpha_1 \alpha_3}^{(1)} G_{\alpha_2 \alpha_4}^{(1)} \\ &\quad + \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(1)} G_{\alpha_2 \alpha_3}^{(1)}\end{aligned}$$

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 &\quad + \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(1)} G_{\alpha_2 \alpha_3}^{(1)} \\
 &= \mathbb{E}\left[z_{i_1;\alpha_1}^{(1)} z_{i_2;\alpha_2}^{(1)}\right] \mathbb{E}\left[z_{i_3;\alpha_3}^{(1)} z_{i_4;\alpha_4}^{(1)}\right] + \mathbb{E}\left[z_{i_1;\alpha_1}^{(1)} z_{i_3;\alpha_3}^{(1)}\right] \mathbb{E}\left[z_{i_2;\alpha_2}^{(1)} z_{i_4;\alpha_4}^{(1)}\right] \\
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Lembrando de discussão anterior, isso significa que

$$\mathbb{E}[z_{i_1;\alpha_1}^{(1)} z_{i_2;\alpha_2}^{(1)} z_{i_3;\alpha_3}^{(1)} z_{i_4;\alpha_4}^{(1)}]_{\text{connected}} = 0$$

de modo que a distribuição é Gaussiana.

Seguindo a mesma ideia:

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de modo que a distribuição é Gaussiana.

De forma similar, podemos verificar que todos os correlatores conexos de ordem mais alta zeram, e portanto, podem ser gerados por uma Gaussiana.

Lembrando, uma distribuição Gaussiana multivariada z de média s e matriz de covariância K pode ser escrita como

$$\begin{aligned} p(z) &= \frac{1}{\sqrt{|2\pi K|}} \exp \left[-\frac{1}{2} \sum_{\mu, \nu=1}^N (z-s)_\mu K^{\mu\nu} (z-s)_\nu \right] \\ &= \frac{1}{\sqrt{|2\pi K|}} \exp \left[-\frac{1}{2} (z-s)^T K^{-1} (z-s) \right] \end{aligned}$$

onde $K^{\mu\nu} = (K^{-1})_{\mu\nu}$.

Para escrever a distribuição de primeira camada, precisamos da inversa de $\delta_{i_1 i_2} G_{(1)}^{\alpha_1 \alpha_2}$, para cada par de índices i_1, i_2 , que satisfaça

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$$\sum_{j=1}^{n_1} \sum_{\beta \in D} (\delta_{i_1 j} G_{(1)}^{\alpha_1 \beta}) (\delta_{i_2 j} G_{\beta \alpha_2}^{(1)}) = \delta_{i_1 i_2} \delta_{\alpha_2}^{\alpha_1}$$

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$G_{(1)}^{\alpha\beta}$ denota as entradas da matriz $(G^{(1)})_{\alpha\beta}^{-1} \in \mathbb{R}^{N_D}$, com as métricas $G_{\alpha\beta}^{(1)}$ para duas entradas x_α, x_β , satisfazendo

$$\sum_{\beta \in D} G_{(1)}^{\alpha_1 \beta} G_{\beta \alpha_2}^{(1)} = \delta_{\alpha_2}^{\alpha_1}$$

Podemos, então, escrever a distribuição $p(z^{(1)}|D)$ na forma:

$$p(z^{(1)}|D) = \frac{1}{Z} e^{-S(z^{(1)})}$$

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$$S(z^{(1)}) = \frac{1}{2} \sum_{i=1}^{n_1} \sum_{\alpha_1, \alpha_2 \in D} z_{i;\alpha_1}^{(1)} G_{(1)}^{\alpha_1 \alpha_2} z_{i;\alpha_2}^{(1)}$$

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$$\left(= \sum_{i,j=1}^{n_1} \sum_{\alpha_1, \alpha_2 \in D} z_{i;\alpha_1}^{(1)} \delta_{ij} G_{(1)}^{\alpha_1 \alpha_2} z_{j;\alpha_2}^{(1)} \right)$$

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e Z é a *função de partição*:

$$Z = \int \left[\prod_{i,\alpha} dz_{i,\alpha}^{(1)} \right] e^{-S(z^{(1)})} = |2\pi G^{(1)}|^{\frac{n_1}{2}}$$

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Como visto no capítulo anterior, podemos expressar a distribuição na forma

$$p(z | D) = \int \left[\prod_j db_j p(b_j) \right] \left[\prod_{j,k} dW_{jk} p(W_{jk}) \right] \prod_{j,\alpha} \delta \left(z_{j;\alpha} - b_i - \sum_j W_{jk} x_{j;\alpha} \right)$$

suprimindo por hora os sobrescritos (1).

Por hipótese:

$$p(b_j) = \frac{1}{\sqrt{2\pi C_b}} \exp\left(-\frac{b_j^2}{2C_b}\right)$$

$$p(W_{jk}) = \frac{1}{\sqrt{2\pi C_W/n_0}} \exp\left(-\frac{n_0 W_{jk}^2}{2C_W}\right)$$

Vamos usar a **transformação de Hubbard-Stratonovich**.
I.e., a representação integral do delta de Dirac:

$$\delta(z - a) = \int \frac{d\Lambda}{2\pi} e^{i\Lambda(z-a)}$$

Obtemos:

$$p(z | D) = \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \exp\left(-\frac{b_j^2}{2C_b}\right) \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \exp\left(-\frac{n_0 W_{jk}^2}{2C_W}\right) \right] \left[\prod_{j,\alpha} \int \frac{d\Lambda}{2\pi} e^{i\Lambda(z-a)} \right]$$

Obtemos:

$$\begin{aligned}
 p(z \mid D) &= \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \exp\left(-\frac{b_j^2}{2C_b}\right) \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \exp\left(-\frac{n_0 W_{jk}^2}{2C_W}\right) \right] \left[\prod_{j,\alpha} \int \frac{d\Lambda}{2\pi} e^{i\Lambda(z-a)} \right] \\
 &= \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \right] \left[\prod_{i,j} \frac{dW_{ij}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{i,\alpha} \frac{d\Lambda_i^\alpha}{2\pi} \right] \\
 &\quad \times \exp \left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{j,k} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha} \right) \right]
 \end{aligned}$$

Completando quadrados:

$$\begin{aligned}
 & \sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{i,j} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{k;\alpha} \right) \\
 &= -\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 - \frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i \frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \\
 & \quad - \frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha}
 \end{aligned}$$

Voltando à exponencial,

$$\int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\ \times \exp \left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{jk} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha} \right) \right]$$

Voltando à exponencial,

$$\begin{aligned}
& \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\
& \times \exp \left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{jk} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha} \right) \right] \\
& = \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\
& \exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 \right] \exp \left[-\frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i \frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \right] \\
& \exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]
\end{aligned}$$

$$\begin{aligned}
&= \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\
&\exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 \right] \exp \left[-\frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i\frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \right] \\
&\exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]
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&= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\
&\exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 \right] \exp \left[-\frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i\frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \right] \\
&\exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right] \\
&\stackrel{\times}{=} \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}} \right] \exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 \right] \\
&\times \int \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \exp \left[-\frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i\frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \right] \\
&\int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]
\end{aligned}$$

Olhando bem, são integrais Gaussianas deslocadas por uma média:

$$\begin{aligned} \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \right] \exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_{\alpha} \Lambda_j^{\alpha} \right)^2 \right] \\ = \prod_j \int \left[\frac{db_j}{\sqrt{2\pi C_b}} \exp \left(-\frac{(b_j - \mu_b)^2}{2C_b} \right) \right] \end{aligned}$$

com a média $\mu_b = -iC_b \sum_{\alpha} \Lambda_j^{\alpha}$. Assim, cada integral dessa é igual a 1, e some da representação.

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com a média $\mu_b = -iC_b \sum_{\alpha} \Lambda_j^{\alpha}$. Assim, cada integral dessa é igual a 1, e some da representação.

De forma análoga, a integral envolvendo W_{jk} vira 1.

Então...

$$\begin{aligned}
 p(z \mid D) &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_{\alpha} \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_{\alpha} \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right] \\
 &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \sum_{\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} x_{k;\alpha_1} x_{k;\alpha_2} - \frac{C_b}{2} \sum_j \sum_{\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right] \\
 &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp \left[-\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} \left(C_b + C_W \frac{1}{n_0} \sum_k x_{k;\alpha_1} x_{k;\alpha_2} \right) + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right] \\
 &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp \left[-\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]
 \end{aligned}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$-\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j; \alpha}$$

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 & -\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j; \alpha} \\
 & = -\frac{1}{2} [\Lambda^T G \Lambda - 2i \Lambda^T z]
 \end{aligned}$$

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 & = -\frac{1}{2} [\Lambda^T G \Lambda - 2i \Lambda^T z] \\
 & = -\frac{1}{2} [\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda]
 \end{aligned}$$

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 & = -\frac{1}{2} [\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda] \\
 & = -\frac{1}{2} [\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i (G^{-1} z)^T G \Lambda - (G^{-1} z)^T G (G^{-1} z) + (G^{-1} z)^T G (G^{-1} z)]
 \end{aligned}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

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 & = -\frac{1}{2} [\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda] \\
 & = -\frac{1}{2} [\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i (G^{-1} z)^T G \Lambda - (G^{-1} z)^T G (G^{-1} z) + (G^{-1} z)^T G (G^{-1} z)] \\
 & = -\frac{1}{2} [\Lambda^T G (\Lambda - i G^{-1} z) - i (G^{-1} z)^T G (\Lambda - G^{-1} z) + z^T G^{-1} z]
 \end{aligned}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$\begin{aligned}
 & -\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j; \alpha} \\
 & = -\frac{1}{2} [\Lambda^T G \Lambda - 2i \Lambda^T z] \\
 & = -\frac{1}{2} [\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda] \\
 & = -\frac{1}{2} [\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i (G^{-1} z)^T G \Lambda - (G^{-1} z)^T G (G^{-1} z) + (G^{-1} z)^T G (G^{-1} z)] \\
 & = -\frac{1}{2} [\Lambda^T G (\Lambda - i G^{-1} z) - i (G^{-1} z)^T G (\Lambda - G^{-1} z) + z^T G^{-1} z] \\
 & = -\frac{1}{2} [(\Lambda - i G^{-1} z)^T G (\Lambda - i G^{-1} z) + z^T G^{-1} z].
 \end{aligned}$$

Ou seja,

$$-\frac{1}{2}\Lambda^T G \Lambda + i\Lambda^T z = -\frac{1}{2} \left[(\Lambda - iG^{-1}z)^T G (\Lambda - iG^{-1}z) + z^T G^{-1}z \right].$$

Em notação de somatório, para ficar igual ao livro

$$\begin{aligned} & -\frac{1}{2} \left[(\Lambda - iG^{-1}z)^T G (\Lambda - iG^{-1}z) + z^T G^{-1}z \right] \\ &= -\frac{1}{2} \sum_{j, \alpha_1, \alpha_2} \left[G_{\alpha_1 \alpha_2}^{(1)} \left(\Lambda_{\alpha_1 j} - i \sum_{\beta_1} G_{(1)}^{\alpha_1 \beta_1} z_{j; \beta_1}^{(1)} \right) \left(\Lambda_{\alpha_2 j} - i \sum_{\beta_2} G_{(1)}^{\alpha_2 \beta_2} z_{j; \beta_2}^{(1)} \right) + G_{(1)}^{\alpha_1 \alpha_2} z_{j; \alpha_1}^{(1)} z_{j; \alpha_2}^{(1)} \right] \end{aligned}$$

Assim,

$$\begin{aligned}
 p(z \mid D) &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp \left[(\Lambda - iG^{-1}z)^T G (\Lambda - iG^{-1}z) \right] \exp \left[z^T G^{-1} z \right] \\
 &= |2\pi G^{-1}|^{\frac{n_1}{2}} \exp \left[z^T G^{-1} z \right] \\
 &= \frac{1}{|2\pi G^{(1)}|^{\frac{n_1}{2}}} \exp \left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2} G_{(1)}^{\alpha_1\alpha_2} z_{j;\alpha_1}^{(1)} z_{j;\alpha_2}^{(1)} \right]
 \end{aligned}$$

voltando ao final o sobrescrito e a notação de somatório.

Assim,

$$\begin{aligned}
 p(z \mid D) &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp \left[(\Lambda - iG^{-1}z)^T G (\Lambda - iG^{-1}z) \right] \exp \left[z^T G^{-1}z \right] \\
 &= |2\pi G^{-1}|^{\frac{n_1}{2}} \exp \left[z^T G^{-1}z \right] \\
 &= \frac{1}{|2\pi G^{(1)}|^{\frac{n_1}{2}}} \exp \left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2} G_{(1)}^{\alpha_1\alpha_2} z_{j;\alpha_1}^{(1)} z_{j;\alpha_2}^{(1)} \right]
 \end{aligned}$$

voltando ao final o sobrescrito e a notação de somatório.

Ou seja, o mesmo termo que encontramos na seção “Wick this way”.

Introdução

4.1 - 1ª Camada: A boa e velha Gaussiana

“Wick this way”, derivação combinatória

”Hubbard-Stratonovich this way”: derivação algébrica via ação

Ação Gaussiana em ação

Seção 4.2 - 2ª Camada: A gênese da não-gaussianidade

Introdução

Distribuição condicional de segunda camada

Wick Wick Wick: derivação combinatória

Schwinger-Dyson this way: derivação algébrica

Ideia: computar dois valores importantes para a seção seguinte:

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- ▶ $\mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right];$

Ideia: computar dois valores importantes para a seção seguinte:

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- ▶ $\mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \sigma \left(z_{i_2; \alpha_3}^{(1)} \right) \sigma \left(z_{i_2; \alpha_4}^{(1)} \right) \right].$

$$\begin{aligned} & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\ &= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \end{aligned}$$

$$\begin{aligned}
& \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\
&= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \times \exp \left(-\frac{1}{2} \sum_{j=1}^{n_1} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{j; \beta_1} z_{j; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2})
\end{aligned}$$

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& \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\
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&= \left\{ \prod_{i \neq i_1} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left[-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i; \beta_1} z_{i; \beta_2} \right] \right\}
\end{aligned}$$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\
 &= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \times \exp \left(-\frac{1}{2} \sum_{j=1}^{n_1} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{j; \beta_1} z_{j; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &= \left\{ \prod_{i \neq i_1} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left[-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i; \beta_1} z_{i; \beta_2} \right] \right\} \\
 &\times \int \left[\prod_{\alpha \in D} \frac{dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2})
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 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\
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 &= \left\{ \prod_{i \neq i_1} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left[-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i; \beta_1} z_{i; \beta_2} \right] \right\} \\
 &\times \int \left[\prod_{\alpha \in D} \frac{dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &= \{1\} \times \left[\int \prod_{\alpha \in D} \frac{dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2})
 \end{aligned}$$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1}^{(1)}; \alpha_1 \right) \sigma \left(z_{i_1}^{(1)}; \alpha_2 \right) \right] \\
 &= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \times \exp \left(-\frac{1}{2} \sum_{j=1}^{n_1} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{j;\beta_1} z_{j;\beta_2} \right) \sigma(z_{i_1}; \alpha_1) \sigma(z_{i_1}; \alpha_2) \\
 &= \left\{ \prod_{i \neq i_1} \int \left[\prod_{\alpha \in D} \frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left[-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i;\beta_1} z_{i;\beta_2} \right] \right\} \\
 &\quad \times \int \left[\prod_{\alpha \in D} \frac{dz_{i_1;\alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i_1;\beta_1} z_{i_1;\beta_2} \right) \sigma(z_{i_1}; \alpha_1) \sigma(z_{i_1}; \alpha_2) \\
 &= \{1\} \times \left[\int \prod_{\alpha \in D} \frac{dz_{i_1;\alpha}}{\sqrt{|2\pi G^{(1)}|}} \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i_1;\beta_1} z_{i_1;\beta_2} \right) \sigma(z_{i_1}; \alpha_1) \sigma(z_{i_1}; \alpha_2) \right] \\
 &:= \langle \sigma(z_{i_1}; \alpha_1) \sigma(z_{i_1}; \alpha_2) \rangle_{G^{(1)}}.
 \end{aligned}$$

Com a expressão do final, reintroduzimos a notação para o correlator de $F(z_{\alpha_1}, \dots, z_{\alpha_m})$ com matriz de variância $g = (g_{\alpha_1 \alpha_2})_{N_D \times N_D}$:

$$\langle F(z_{\alpha_1}, \dots, z_{\alpha_m}) \rangle_g \equiv \int \left[\frac{\prod_{\alpha \in D} dz_{\alpha}}{\sqrt{|2\pi g|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} g^{\beta_1 \beta_2} z_{\beta_1} z_{\beta_2} \right) F(z_{\alpha_1}, \dots, z_{\alpha_m})$$

Com a expressão do final, reintroduzimos a notação para o correlator de $F(z_{\alpha_1}, \dots, z_{\alpha_m})$ com matriz de variância $g = (g_{\alpha_1 \alpha_2})_{N_D \times N_D}$:

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e denotamos ainda:

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e denotamos ainda:

$$\sigma_{\alpha} := \sigma(z_{\alpha})$$

Assim, podemos escrever

$$\mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] = \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \rangle_{G^{(1)}}$$

É possível ainda generalizar para correlatores de mais pontos.
Para todas as saídas do mesmo neurônio i_1 :

$$\mathbb{E} \left[\sigma(z_{i_1; \alpha_1}^{(1)}) \sigma(z_{i_1; \alpha_2}^{(1)}) \sigma(z_{i_1; \alpha_3}^{(1)}) \sigma(z_{i_1; \alpha_4}^{(1)}) \right] = \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \sigma_{\alpha_3} \sigma_{\alpha_4} \rangle_{G^{(1)}}$$

fazendo as exatas mesmas etapas.

Para $i_1 \neq i_2$

$$\mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \sigma \left(z_{i_2; \alpha_3}^{(1)} \right) \sigma \left(z_{i_2; \alpha_4}^{(1)} \right) \right]$$

Para $i_1 \neq i_2$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \sigma \left(z_{i_2; \alpha_3}^{(1)} \right) \sigma \left(z_{i_2; \alpha_4}^{(1)} \right) \right] \\
 &= \left\{ \prod_{i \notin \{i_1, i_2\}} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i; \beta_1} z_{i; \beta_2} \right) \right\} \\
 &\times \int \left[\frac{\prod_{\alpha \in D} dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &\times \int \left[\frac{\prod_{\alpha \in D} dz_{i_2; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_2; \beta_1} z_{i_2; \beta_2} \right) \sigma(z_{i_2; \alpha_3}) \sigma(z_{i_2; \alpha_4})
 \end{aligned}$$

Para $i_1 \neq i_2$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \sigma \left(z_{i_2; \alpha_3}^{(1)} \right) \sigma \left(z_{i_2; \alpha_4}^{(1)} \right) \right] \\
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 &= \langle \sigma(z_{\alpha_1}) \sigma(z_{\alpha_2}) \rangle_{G^{(1)}} \langle \sigma(z_{\alpha_3}) \sigma(z_{\alpha_4}) \rangle_{G^{(1)}}
 \end{aligned}$$

I.e.:

$$\mathbb{E} \left[\sigma \left(z_{i_1}^{(1)}; \alpha_1 \right) \sigma \left(z_{i_1}^{(1)}; \alpha_2 \right) \sigma \left(z_{i_2}^{(1)}; \alpha_3 \right) \sigma \left(z_{i_2}^{(1)}; \alpha_4 \right) \right] = \langle \sigma(z_{\alpha_1}) \sigma(z_{\alpha_2}) \rangle_{G(1)} \langle \sigma(z_{\alpha_3}) \sigma(z_{\alpha_4}) \rangle_{G(1)}$$

Isso ilustra a independência entre as distribuições de neurônios diferentes.

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“Wick this way”, derivação combinatória

”Hubbard-Stratonovich this way”: derivação algébrica via ação

Ação Gaussiana em ação

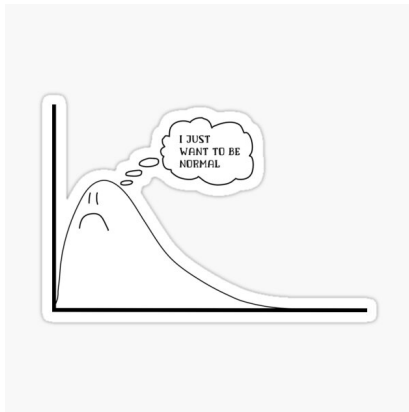
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Lembrando, denotamos:

$$\sigma_{i;\alpha}^{(1)} := \sigma(z_{i;\alpha}^{(1)}). \quad (1)$$

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As entradas da segunda camada são dadas por:

$$z_{i,\alpha}^{(2)} \equiv z_i^{(2)}(x_\alpha) = b_i^{(2)} + \sum_{j=1}^{n_1} W_{ij}^{(2)} \sigma_{j;\alpha}^{(1)}, \quad \text{para } i = 1, \dots, n_2. \quad (2)$$

A distribuição conjunta da segunda e primeira camada pode ser dada por::

$$p(z^{(2)}, z^{(1)} \mid D) = p(z^{(2)} \mid z^{(1)})p(z^{(1)} \mid D) \quad (3)$$

1º: $p(z^{(1)} \mid D)$ - Seção 4.1;

A distribuição conjunta da segunda e primeira camada pode ser dada por::

$$p(z^{(2)}, z^{(1)} | D) = p(z^{(2)} | z^{(1)})p(z^{(1)} | D) \quad (3)$$

1º: $p(z^{(1)} | D)$ - Seção 4.1;

2º: Como definido na seção 2:

$$p(z^{(2)} | z^{(1)}) = \int \left[\prod_i db_i^{(2)} p(b_i^{(2)}) \right] \left[\prod_{i,j} dW_{ij}^{(2)} p(W_{ij}^{(2)}) \right] \prod_{i,\alpha} \delta \left(z_i^{(2)}(x_\alpha) - b_i^{(2)} - \sum_{j=1}^{n_1} W_{ij}^{(2)} \sigma_{j;\alpha}^{(1)} \right) \quad (4)$$

Ainda, podemos calcular as probabilidades marginalizadas como:

$$p(z^{(2)} | D) = \int \left[\prod_{i;\alpha} dz_{i;\alpha}^{(1)} \right] p(z^{(1)} | z^{(2)}) p(z^{(1)} | D). \quad (5)$$

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Podemos calcular $p(z^{(2)} | z^{(1)})$ da exata mesma forma que $p(z^{(1)} | D)$:

- ▶ índices $1 \rightarrow 2$;
- ▶ entradas $x_{j;\alpha} \rightarrow \sigma_{j;\alpha}^{(1)}$.

Desse modo, a probabilidade fica:

$$p(z^{(2)} | z^{(1)}) = \frac{1}{|2\pi \widehat{G}^{(2)}|^{\frac{n_2}{2}}} \exp \left[-\frac{1}{2} \sum_{j, \alpha_1, \alpha_2}^{n_2} \widehat{G}_{(2)}^{\alpha_1 \alpha_2} z_{j;\alpha_1}^{(2)} z_{j;\alpha_2}^{(2)} \right] \quad (6)$$

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onde definimos a **métrica estocástica de segunda camada**:

$$\widehat{G}_{\alpha_1 \alpha_2}^{(2)} := C_b^{(2)} + C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} \sigma_{j; \alpha_1} \sigma_{j; \alpha_2} \quad (7)$$

Importante ressaltar que a variável $\widehat{G}^{(2)}$ é *estocástica*:

$$\widehat{G}_{\alpha_1\alpha_2}^{(2)} := C_b^{(2)} + C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} \sigma_{j;\alpha_1} \sigma_{j;\alpha_2} \quad (8)$$

pois depende da variável aleatória $\sigma^{(1)} = \sigma(z^{(1)})$.

Podemos então calcular seu valor esperado:

$$\mathbb{E}[\widehat{G}_{\alpha_1\alpha_2}^{(2)}] = C_b^{(2)} + C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} \mathbb{E}[\sigma_{j;\alpha_1} \sigma_{j;\alpha_2}]$$

Podemos então calcular seu valor esperado:

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$$\begin{aligned}\mathbb{E}[\widehat{G}_{\alpha_1\alpha_2}^{(2)}] &= C_b^{(2)} + C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} \mathbb{E}[\sigma_{j;\alpha_1} \sigma_{j;\alpha_2}] \\ &= C_b^{(2)} + C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} \langle \sigma_{j;\alpha_1} \sigma_{j;\alpha_2} \rangle_{G^{(1)}} \\ &:= G_{\alpha_1\alpha_2}^{(2)}.\end{aligned}$$

Chamamos o valor $G_{\alpha_1\alpha_2}^{(2)}$ de **métrica média de 2^{a} camada**.

Definimos ainda **flutuação de 2^a camada**:

$$\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)} := \widehat{G}_{\alpha_1 \alpha_2}^{(2)} - G_{\alpha_1 \alpha_2}^{(2)}$$

Definimos ainda **flutuação de 2ª camada**:

$$\begin{aligned}\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)} &:= \widehat{G}_{\alpha_1 \alpha_2}^{(2)} - G_{\alpha_1 \alpha_2}^{(2)} \\ &= C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} (\sigma_{j; \alpha_1}^{(1)} \sigma_{j; \alpha_2}^{(1)} - \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \rangle_{G^{(1)}})\end{aligned}$$

Definimos ainda **flutuação de 2ª camada**:

$$\begin{aligned}\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)} &:= \widehat{G}_{\alpha_1 \alpha_2}^{(2)} - G_{\alpha_1 \alpha_2}^{(2)} \\ &= C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} (\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} - \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \rangle_{G^{(1)}})\end{aligned}$$

Note que por construção, $\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)}$ tem média 0 sobre a primeira camada de ativações:

$$\mathbb{E}[\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)}] = \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)}] - \mathbb{E}[\mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)}]] = 0$$

Estamos interessados no correlator de 2 pontos de $\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)}$:

$$\mathbb{E} \left[\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)} \widehat{\Delta G}_{\alpha_3 \alpha_4}^{(2)} \right]$$

Estamos interessados no correlator de 2 pontos de $\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)}$:

$$\begin{aligned} & \mathbb{E} \left[\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)} \widehat{\Delta G}_{\alpha_3 \alpha_4}^{(2)} \right] \\ &= \left(\frac{C_W^{(2)}}{n_1} \right)^2 \sum_{j,k=1}^{n_1} \mathbb{E} \left[\left(\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} - \mathbb{E}[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)}] \right) \left(\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} - \mathbb{E}[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)}] \right) \right] \end{aligned}$$

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Temos:

$$\mathbb{E} \left[\left(\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} - \mathbb{E}[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)}] \right) \left(\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} - \mathbb{E}[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)}] \right) \right]$$

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$$\begin{aligned} & \mathbb{E} \left[\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)} \widehat{\Delta G}_{\alpha_3 \alpha_4}^{(2)} \right] \\ &= \left(\frac{C_W^{(2)}}{n_1} \right)^2 \sum_{j,k=1}^{n_1} \mathbb{E} \left[\left(\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} - \mathbb{E}[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)}] \right) \left(\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} - \mathbb{E}[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)}] \right) \right] \end{aligned}$$

Temos:

$$\begin{aligned} & \mathbb{E} \left[\left(\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} - \mathbb{E}[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)}] \right) \left(\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} - \mathbb{E}[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)}] \right) \right] \\ &= \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} \right] - \mathbb{E} \left[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} \right] \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \right] \\ & \quad - \mathbb{E} \left[\sigma_{j;\alpha_3}^{(1)} \sigma_{j;\alpha_4}^{(1)} \right] \mathbb{E} \left[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} \right] + \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \right] \mathbb{E} \left[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} \right] \end{aligned}$$

Estamos interessados no correlator de 2 pontos de $\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)}$:

$$\begin{aligned} & \mathbb{E} \left[\widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)} \widehat{\Delta G}_{\alpha_3 \alpha_4}^{(2)} \right] \\ &= \left(\frac{C_W^{(2)}}{n_1} \right)^2 \sum_{j,k=1}^{n_1} \mathbb{E} \left[\left(\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} - \mathbb{E}[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)}] \right) \left(\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} - \mathbb{E}[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)}] \right) \right] \end{aligned}$$

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Note que, para $j \neq k$,

$$\mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} \right] = \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \right] \mathbb{E} \left[\sigma_{j;\alpha_3}^{(1)} \sigma_{j;\alpha_4}^{(1)} \right]$$

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Com isso,

$$\left(\frac{C_W^{(2)}}{n_1} \right)^2 \sum_{j,k=1}^{n_1} \mathbb{E} \left[\left(\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} - \mathbb{E}[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)}] \right) \left(\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} - \mathbb{E}[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)}] \right) \right]$$

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Com isso,

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$$\mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} \right] = \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \right] \mathbb{E} \left[\sigma_{j;\alpha_3}^{(1)} \sigma_{j;\alpha_4}^{(1)} \right]$$

Com isso,

$$\begin{aligned} & \left(\frac{C_W^{(2)}}{n_1} \right)^2 \sum_{j,k=1}^{n_1} \mathbb{E} \left[\left(\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} - \mathbb{E}[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)}] \right) \left(\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} - \mathbb{E}[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)}] \right) \right] \\ &= \left(\frac{C_W^{(2)}}{n_1} \right)^2 \sum_{j,k=1}^{n_1} \left\{ \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} \right] - \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \right] \mathbb{E} \left[\sigma_{k;\alpha_3}^{(1)} \sigma_{k;\alpha_4}^{(1)} \right] \right\} \\ &= \left(\frac{C_W^{(2)}}{n_1} \right)^2 \sum_{j=1}^{n_1} \left\{ \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \sigma_{j;\alpha_3}^{(1)} \sigma_{j;\alpha_4}^{(1)} \right] - \mathbb{E} \left[\sigma_{j;\alpha_1}^{(1)} \sigma_{j;\alpha_2}^{(1)} \right] \mathbb{E} \left[\sigma_{j;\alpha_3}^{(1)} \sigma_{j;\alpha_4}^{(1)} \right] \right\} \end{aligned}$$

Note que, para $j \neq k$,

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Com isso,

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O valor

$$V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} := \left(C_W^{(2)}\right)^2 [\langle \sigma_{\alpha_1} \sigma_{\alpha_2} \sigma_{\alpha_3} \sigma_{\alpha_4} \rangle_{G^{(1)}} - \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \rangle_{G^{(1)}} \langle \sigma_{\alpha_3} \sigma_{\alpha_4} \rangle_{G^{(1)}}]$$

é chamado de **vértice de quatro pontos** de 2^a camada.

O valor

$$V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} := \left(C_W^{(2)}\right)^2 [\langle \sigma_{\alpha_1} \sigma_{\alpha_2} \sigma_{\alpha_3} \sigma_{\alpha_4} \rangle_{G^{(1)}} - \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \rangle_{G^{(1)}} \langle \sigma_{\alpha_3} \sigma_{\alpha_4} \rangle_{G^{(1)}}]$$

é chamado de **vértice de quatro pontos** de 2^{a} camada. Desse modo, podemos escrever

$$\mathbb{E} \left[\widehat{\Delta G}_{\alpha_1\alpha_2}^{(2)} \widehat{\Delta G}_{\alpha_3\alpha_4}^{(2)} \right] = \frac{1}{n_1} V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)}.$$

Observações:

- ▶ $V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} == V(x_{\alpha_1}, x_{\alpha_2}; x_{\alpha_3}, x_{\alpha_4});$
- ▶ O vértice de quatro pontos é simétrico sobre $\alpha_1 \leftrightarrow \alpha_2,$
 $\alpha_3 \leftrightarrow \alpha_4, (\alpha_1, \alpha_2) \leftrightarrow (\alpha_3, \alpha_4).$

Observações:

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 - ▶ $V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} = V_{(\alpha_2\alpha_1)(\alpha_3\alpha_4)}^{(2)};$
 - ▶ $V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} = V_{(\alpha_1\alpha_2)(\alpha_4\alpha_3)}^{(2)};$
 - ▶ $V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} = V_{(\alpha_3\alpha_4)(\alpha_1\alpha_2)}^{(2)};$

Observações:

- ▶ $V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} = V(x_{\alpha_1}, x_{\alpha_2}; x_{\alpha_3}, x_{\alpha_4});$
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 - ▶ $V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} = V_{(\alpha_2\alpha_1)(\alpha_3\alpha_4)}^{(2)};$
 - ▶ $V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} = V_{(\alpha_1\alpha_2)(\alpha_4\alpha_3)}^{(2)};$
 - ▶ $V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} = V_{(\alpha_3\alpha_4)(\alpha_1\alpha_2)}^{(2)};$
- ▶ Quando $n_1 \gg 1$, $\frac{1}{n_1} V_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}^{(2)} \rightarrow 0$, de modo que a distribuição se aproxima de uma Gaussiana.

Introdução

4.1 - 1ª Camada: A boa e velha Gaussiana

“Wick this way”, derivação combinatória

”Hubbard-Stratonovich this way”: derivação algébrica via ação

Ação Gaussiana em ação

Seção 4.2 - 2ª Camada: A gênese da não-gaussianidade

Introdução

Distribuição condicional de segunda camada

Wick Wick Wick: derivação combinatória

Schwinger-Dyson this way: derivação algébrica

Como visto na Equação (6), $p(z^{(2)} | z^{(1)})$ é gaussiana (em relação a $z^{(1)}$), com média 0 e variância (estocástica) $\widehat{G}_{\alpha_1\alpha_2}^{(2)}$:

$$p(z^{(2)} | z^{(1)}) = \frac{1}{|2\pi\widehat{G}^{(2)}|^{\frac{n_2}{2}}} \exp\left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2}^{n_2} \widehat{G}_{(2)}^{\alpha_1\alpha_2} z_{j;\alpha_1}^{(2)} z_{j;\alpha_2}^{(2)}\right]$$

Como visto na Equação (6), $p(z^{(2)} | z^{(1)})$ é gaussiana (em relação a $z^{(1)}$), com média 0 e variância (estocástica) $\widehat{G}_{\alpha_1\alpha_2}^{(2)}$:

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Isso nos permite usar as *contrações de Wick*.

Como visto na Equação (6), $p(z^{(2)} | z^{(1)})$ é gaussiana (em relação a $z^{(1)}$), com média 0 e variância (estocástica) $\widehat{G}_{\alpha_1\alpha_2}^{(2)}$:

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Isso nos permite usar as *contrações de Wick*. Para o correlator de 2 pontos:

$$\begin{aligned} \mathbb{E}\left[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)}\right] &= \delta_{i_1 i_2} \mathbb{E}[\widehat{G}_{\alpha_1\alpha_2}^{(2)}] \\ &= \delta_{i_1 i_2} G_{\alpha_1\alpha_2}^{(2)} \\ &= \delta_{i_1 i_2} \left(C_b^{(2)} + C_W^{(2)} \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \rangle_{G^{(1)}}\right) \end{aligned}$$

Como visto na Equação (6), $p(z^{(2)} | z^{(1)})$ é gaussiana (em relação a $z^{(1)}$), com média 0 e variância (estocástica) $\widehat{G}_{\alpha_1\alpha_2}^{(2)}$:

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Obs: o livro não explica, mas o valor esperado deve aparecer aqui no teorema de Wick pela variância ser uma variável estocástica. Uma conta parecida com a Seção 4.1 (Wick this way) também revela onde surge a esperança.

No correlator de 4 pontos:

$$\begin{aligned} & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\ &= \delta_{i_1 i_2} \delta_{i_3 i_4} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)} \widehat{G}_{\alpha_3 \alpha_4}^{(2)}] + \delta_{i_1 i_3} \delta_{i_2 i_4} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_3}^{(2)} \widehat{G}_{\alpha_2 \alpha_4}^{(2)}] + \delta_{i_1 i_4} \delta_{i_2 i_3} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_4}^{(2)} \widehat{G}_{\alpha_2 \alpha_3}^{(2)}] \end{aligned}$$

No correlator de 4 pontos:

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Lembrando,

$$\blacktriangleright \widehat{G}_{\alpha_1 \alpha_2}^{(2)} = G_{\alpha_1 \alpha_2}^{(2)} + \widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)};$$

No correlator de 4 pontos:

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- ▶ $\widehat{G}_{\alpha_1 \alpha_2}^{(2)} = G_{\alpha_1 \alpha_2}^{(2)} + \widehat{\Delta G}_{\alpha_1 \alpha_2}^{(2)};$
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$$\begin{aligned} & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\ &= \delta_{i_1 i_2} \delta_{i_3 i_4} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)} \widehat{G}_{\alpha_3 \alpha_4}^{(2)}] + \delta_{i_1 i_3} \delta_{i_2 i_4} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_3}^{(2)} \widehat{G}_{\alpha_2 \alpha_4}^{(2)}] + \delta_{i_1 i_4} \delta_{i_2 i_3} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_4}^{(2)} \widehat{G}_{\alpha_2 \alpha_3}^{(2)}] \end{aligned}$$

Lembrando,

- ▶ $\widehat{G}_{\alpha_1 \alpha_2}^{(2)} = G_{\alpha_1 \alpha_2}^{(2)} + \overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)};$
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- ▶ $\mathbb{E}[\overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)}] = 0;$

Assim:

$$\mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)} \widehat{G}_{\alpha_3 \alpha_4}^{(2)}] = \mathbb{E}[(G_{\alpha_1 \alpha_2}^{(2)} + \overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)})(G_{\alpha_3 \alpha_4}^{(2)} + \overline{\Delta G}_{\alpha_3 \alpha_4}^{(2)})]$$

No correlator de 4 pontos:

$$\begin{aligned} & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\ &= \delta_{i_1 i_2} \delta_{i_3 i_4} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)} \widehat{G}_{\alpha_3 \alpha_4}^{(2)}] + \delta_{i_1 i_3} \delta_{i_2 i_4} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_3}^{(2)} \widehat{G}_{\alpha_2 \alpha_4}^{(2)}] + \delta_{i_1 i_4} \delta_{i_2 i_3} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_4}^{(2)} \widehat{G}_{\alpha_2 \alpha_3}^{(2)}] \end{aligned}$$

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Assim:

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- ▶ $\mathbb{E}[\overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)}] = 0$;

Assim:

$$\begin{aligned} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)} \widehat{G}_{\alpha_3 \alpha_4}^{(2)}] &= \mathbb{E}[(G_{\alpha_1 \alpha_2}^{(2)} + \overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)})(G_{\alpha_3 \alpha_4}^{(2)} + \overline{\Delta G}_{\alpha_3 \alpha_4}^{(2)})] \\ &= G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} + G_{\alpha_1 \alpha_2}^{(2)} \mathbb{E}[\overline{\Delta G}_{\alpha_3 \alpha_4}^{(2)}] + G_{\alpha_3 \alpha_4}^{(2)} \mathbb{E}[\overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)}] + \mathbb{E}[\overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)} \overline{\Delta G}_{\alpha_3 \alpha_4}^{(2)}] \\ &= G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} + V_{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)}^{(2)} \end{aligned}$$

No correlator de 4 pontos:

$$\begin{aligned} & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\ &= \delta_{i_1 i_2} \delta_{i_3 i_4} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)} \widehat{G}_{\alpha_3 \alpha_4}^{(2)}] + \delta_{i_1 i_3} \delta_{i_2 i_4} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_3}^{(2)} \widehat{G}_{\alpha_2 \alpha_4}^{(2)}] + \delta_{i_1 i_4} \delta_{i_2 i_3} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_4}^{(2)} \widehat{G}_{\alpha_2 \alpha_3}^{(2)}] \end{aligned}$$

Lembrando,

- ▶ $\widehat{G}_{\alpha_1 \alpha_2}^{(2)} = G_{\alpha_1 \alpha_2}^{(2)} + \overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)}$;
- ▶ $\mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)}] := G_{\alpha_1 \alpha_2}^{(2)}$;
- ▶ $\mathbb{E}[\overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)}] = 0$;

Assim:

$$\begin{aligned} \mathbb{E}[\widehat{G}_{\alpha_1 \alpha_2}^{(2)} \widehat{G}_{\alpha_3 \alpha_4}^{(2)}] &= \mathbb{E}[(G_{\alpha_1 \alpha_2}^{(2)} + \overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)})(G_{\alpha_3 \alpha_4}^{(2)} + \overline{\Delta G}_{\alpha_3 \alpha_4}^{(2)})] \\ &= G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} + G_{\alpha_1 \alpha_2}^{(2)} \mathbb{E}[\overline{\Delta G}_{\alpha_3 \alpha_4}^{(2)}] + G_{\alpha_3 \alpha_4}^{(2)} \mathbb{E}[\overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)}] + \mathbb{E}[\overline{\Delta G}_{\alpha_1 \alpha_2}^{(2)} \overline{\Delta G}_{\alpha_3 \alpha_4}^{(2)}] \\ &= G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} + V_{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)}^{(2)} \end{aligned}$$

O mesmo argumento se repete para as outras permutações de índices.

Com isso,

$$\begin{aligned} & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\ &= \delta_{i_1 i_2} \delta_{i_3 i_4} \mathbb{E}[\hat{G}_{\alpha_1 \alpha_2}^{(2)} \hat{G}_{\alpha_3 \alpha_4}^{(2)}] + \delta_{i_1 i_3} \delta_{i_2 i_4} \mathbb{E}[\hat{G}_{\alpha_1 \alpha_3}^{(2)} \hat{G}_{\alpha_2 \alpha_4}^{(2)}] + \delta_{i_1 i_4} \delta_{i_2 i_3} \mathbb{E}[\hat{G}_{\alpha_1 \alpha_4}^{(2)} \hat{G}_{\alpha_2 \alpha_3}^{(2)}] \end{aligned}$$

Com isso,

$$\begin{aligned}
 & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
 &= \delta_{i_1 i_2} \delta_{i_3 i_4} \mathbb{E}[\hat{G}_{\alpha_1 \alpha_2}^{(2)} \hat{G}_{\alpha_3 \alpha_4}^{(2)}] + \delta_{i_1 i_3} \delta_{i_2 i_4} \mathbb{E}[\hat{G}_{\alpha_1 \alpha_3}^{(2)} \hat{G}_{\alpha_2 \alpha_4}^{(2)}] + \delta_{i_1 i_4} \delta_{i_2 i_3} \mathbb{E}[\hat{G}_{\alpha_1 \alpha_4}^{(2)} \hat{G}_{\alpha_2 \alpha_3}^{(2)}] \\
 &= \delta_{i_1 i_2} \delta_{i_3 i_4} G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} + \delta_{i_1 i_3} \delta_{i_2 i_4} G_{\alpha_1 \alpha_3}^{(2)} G_{\alpha_2 \alpha_4}^{(2)} + \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(2)} G_{\alpha_2 \alpha_3}^{(2)} \\
 &+ \frac{1}{n_1} [\delta_{i_1 i_2} \delta_{i_3 i_4} V_{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)}^{(2)} + \delta_{i_1 i_3} \delta_{i_2 i_4} V_{(\alpha_1 \alpha_3)(\alpha_2 \alpha_4)}^{(2)} + \delta_{i_1 i_4} \delta_{i_2 i_3} V_{(\alpha_1 \alpha_4)(\alpha_2 \alpha_3)}^{(2)}]
 \end{aligned}$$

Isso nos permite ver que:

$$\begin{aligned}
 & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}]_{\text{connected}} \\
 &= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
 &= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)}] \mathbb{E}[z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_3;\alpha_3}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_4;\alpha_4}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)}]
 \end{aligned}$$

Isso nos permite ver que:

$$\begin{aligned}
 & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}]_{\text{connected}} \\
 &= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
 &- \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)}] \mathbb{E}[z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_3;\alpha_3}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_4;\alpha_4}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)}] \\
 &= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
 &- \delta_{i_1 i_2} \delta_{i_3 i_4} G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} - \delta_{i_1 i_3} \delta_{i_2 i_4} G_{\alpha_1 \alpha_3}^{(2)} G_{\alpha_2 \alpha_4}^{(2)} - \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(2)} G_{\alpha_2 \alpha_3}^{(2)}
 \end{aligned}$$

Isso nos permite ver que:

$$\begin{aligned}
 & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}]_{\text{connected}} \\
 &= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
 &- \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)}] \mathbb{E}[z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_3;\alpha_3}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_4;\alpha_4}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)}] \\
 &= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
 &- \delta_{i_1 i_2} \delta_{i_3 i_4} G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} - \delta_{i_1 i_3} \delta_{i_2 i_4} G_{\alpha_1 \alpha_3}^{(2)} G_{\alpha_2 \alpha_4}^{(2)} - \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(2)} G_{\alpha_2 \alpha_3}^{(2)} \\
 &= \frac{1}{n_1} [\delta_{i_1 i_2} \delta_{i_3 i_4} V_{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)}^{(2)} + \delta_{i_1 i_3} \delta_{i_2 i_4} V_{(\alpha_1 \alpha_3)(\alpha_2 \alpha_4)}^{(2)} + \delta_{i_1 i_4} \delta_{i_2 i_3} V_{(\alpha_1 \alpha_4)(\alpha_2 \alpha_3)}^{(2)}]
 \end{aligned}$$

Isso nos permite ver que:

$$\begin{aligned}
& \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}]_{\text{connected}} \\
&= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
&- \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)}] \mathbb{E}[z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_3;\alpha_3}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_4;\alpha_4}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)}] \\
&= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
&- \delta_{i_1 i_2} \delta_{i_3 i_4} G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} - \delta_{i_1 i_3} \delta_{i_2 i_4} G_{\alpha_1 \alpha_3}^{(2)} G_{\alpha_2 \alpha_4}^{(2)} - \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(2)} G_{\alpha_2 \alpha_3}^{(2)} \\
&= \frac{1}{n_1} [\delta_{i_1 i_2} \delta_{i_3 i_4} V_{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)}^{(2)} + \delta_{i_1 i_3} \delta_{i_2 i_4} V_{(\alpha_1 \alpha_3)(\alpha_2 \alpha_4)}^{(2)} + \delta_{i_1 i_4} \delta_{i_2 i_3} V_{(\alpha_1 \alpha_4)(\alpha_2 \alpha_3)}^{(2)}]
\end{aligned}$$

Observações:

- ▶ Isso evidencia a importância de $V_{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)}^{(2)}$ na “não-gaussianidade” de $p(z^{(2)} | D)$;

Isso nos permite ver que:

$$\begin{aligned}
 & \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}]_{\text{connected}} \\
 &= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
 &- \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)}] \mathbb{E}[z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_3;\alpha_3}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_4;\alpha_4}^{(2)}] - \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_4;\alpha_4}^{(2)}] \mathbb{E}[z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)}] \\
 &= \mathbb{E}[z_{i_1;\alpha_1}^{(2)} z_{i_2;\alpha_2}^{(2)} z_{i_3;\alpha_3}^{(2)} z_{i_4;\alpha_4}^{(2)}] \\
 &- \delta_{i_1 i_2} \delta_{i_3 i_4} G_{\alpha_1 \alpha_2}^{(2)} G_{\alpha_3 \alpha_4}^{(2)} - \delta_{i_1 i_3} \delta_{i_2 i_4} G_{\alpha_1 \alpha_3}^{(2)} G_{\alpha_2 \alpha_4}^{(2)} - \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(2)} G_{\alpha_2 \alpha_3}^{(2)} \\
 &= \frac{1}{n_1} \left[\delta_{i_1 i_2} \delta_{i_3 i_4} V_{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)}^{(2)} + \delta_{i_1 i_3} \delta_{i_2 i_4} V_{(\alpha_1 \alpha_3)(\alpha_2 \alpha_4)}^{(2)} + \delta_{i_1 i_4} \delta_{i_2 i_3} V_{(\alpha_1 \alpha_4)(\alpha_2 \alpha_3)}^{(2)} \right]
 \end{aligned}$$

Observações:

- ▶ Isso evidencia a importância de $V_{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)}^{(2)}$ na “não-gaussianidade” de $p(z^{(2)} | D)$;
- ▶ Mais uma vez mostra que se $n_1 \rightarrow \infty$, o correlator se aproxima de 0 e $p(z^{(2)} | D)$ se aproxima de uma gaussiana.

Vamos atrás de uma ação quártica:

$$S[z] = \frac{1}{2} \sum_{\alpha_1 \alpha_2 \in D} g^{\alpha_1 \alpha_2} \sum_{i=1}^n z_{i;\alpha_1} z_{i;\alpha_2} \\ - \frac{1}{8} \sum_{\alpha_1, \dots, \alpha_4 \in D} v^{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)} \sum_{i_1, i_2=1}^n z_{i_1;\alpha_1} z_{i_1;\alpha_2} z_{i_2;\alpha_3} z_{i_2;\alpha_4}$$

com g e v por hora indeterminados.

Em 4.1, lembramos a notação

$$\langle F(z_{\alpha_1}, \dots, z_{\alpha_m}) \rangle_g \equiv \int \left[\frac{\prod_{\alpha \in D} dz_{\alpha}}{\sqrt{|2\pi g|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} g^{\beta_1 \beta_2} z_{\beta_1} z_{\beta_2} \right) F(z_{\alpha_1}, \dots, z_{\alpha_m})$$

Em 4.1, lembramos a notação

$$\langle F(z_{\alpha_1}, \dots, z_{\alpha_m}) \rangle_g \equiv \int \left[\frac{\prod_{\alpha \in D} dz_{\alpha}}{\sqrt{|2\pi g|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} g^{\beta_1 \beta_2} z_{\beta_1} z_{\beta_2} \right) F(z_{\alpha_1}, \dots, z_{\alpha_m})$$

Agora, introduzimos a notação:

$$\begin{aligned} & \langle\langle F(z_{i_1; \alpha_1}, \dots, z_{i_m; \alpha_m}) \rangle\rangle_g \\ &= \left[\prod_{i=1}^n \frac{\prod_{\alpha \in D} dz_{i; \alpha}}{\sqrt{|2\pi g|}} \right] \exp \left(-\frac{1}{2} \sum_{j=1}^n \sum_{\beta_1, \beta_2 \in D} g^{(\beta_1 \beta_2)} z_{j; \beta_1} z_{j; \beta_2} \right) F(z_{i_1; \alpha_1}, \dots, z_{i_m; \alpha_m}) \end{aligned}$$

Obtemos então:

$$\begin{aligned} & \mathbb{E}[F(z_{i_1;\alpha_1}, \dots, z_{i_m;\alpha_m})] \\ &= \frac{\int \Pi_{i;\alpha} dz_{i;\alpha} e^{-S(z)} F(z_{i_1;\alpha_1}, \dots, z_{i_m;\alpha_m})}{\int \Pi_{i;\alpha} dz_{i;\alpha} e^{-S(z)}} \end{aligned}$$

Obtemos então:

$$\begin{aligned}
 & \mathbb{E}[F(z_{i_1}\alpha_1, \dots, z_{i_m}\alpha_m)] \\
 &= \frac{\int \Pi_{i;\alpha} dz_{i;\alpha} e^{-S(z)} F(z_{i_1;\alpha_1}, \dots, z_{i_m;\alpha_m})}{\int \Pi_{i;\alpha} dz_{i;\alpha} e^{-S(z)}} \\
 &= \frac{\langle\langle \exp \left\{ \frac{1}{8} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1, j_2=1}^n z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} \right\} F(z_{i_1}\alpha_1, \dots, z_{i_m}\alpha_m) \rangle\rangle_g}{\langle\langle \exp \left\{ \frac{1}{8} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1, j_2=1}^n z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} \right\} \rangle\rangle_g}
 \end{aligned}$$

Obtemos então:

$$\begin{aligned}
& \mathbb{E}[F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m})] \\
&= \frac{\int \Pi_{i;\alpha} dz_{i;\alpha} e^{-S(z)} F(z_{i_1;\alpha_1}, \dots, z_{i_m;\alpha_m})}{\int \Pi_{i;\alpha} dz_{i;\alpha} e^{-S(z)}} \\
&= \frac{\langle\langle \exp \left\{ \frac{1}{8} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1, j_2=1}^n z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} \right\} F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m}) \rangle\rangle_g}{\langle\langle \exp \left\{ \frac{1}{8} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1, j_2=1}^n z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} \right\} \rangle\rangle_g} \\
&\langle\langle F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m}) \rangle\rangle_g \\
&+ \frac{1}{8} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1, j_2=1}^n [\langle\langle z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m}) \rangle\rangle_g \\
&- \langle\langle z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} \rangle\rangle_g \langle\langle F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m}) \rangle\rangle_g] \\
&+ O(v^2)
\end{aligned}$$

Obtemos então:

$$\begin{aligned}
& \mathbb{E}[F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m})] \\
&= \frac{\int \prod_{i;\alpha} dz_{i;\alpha} e^{-S(z)} F(z_{i_1;\alpha_1}, \dots, z_{i_m;\alpha_m})}{\int \prod_{i;\alpha} dz_{i;\alpha} e^{-S(z)}} \\
&= \frac{\langle\langle \exp \left\{ \frac{1}{8} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1, j_2=1}^n z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} \right\} F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m}) \rangle\rangle_g}{\langle\langle \exp \left\{ \frac{1}{8} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1, j_2=1}^n z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} \right\} \rangle\rangle_g} \\
&\langle\langle F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m}) \rangle\rangle_g \\
&+ \frac{1}{8} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1, j_2=1}^n [\langle\langle z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m}) \rangle\rangle_g \\
&- \langle\langle z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} \rangle\rangle_g \langle\langle F(z_{i_1\alpha_1}, \dots, z_{i_m\alpha_m}) \rangle\rangle_g] \\
&+ O(v^2)
\end{aligned}$$

Obs: no final, usando a expansão de Taylor $e^v \approx 1 + v + O(v^2)$.

Por exemplo, no correlator de dois pontos,

$$\begin{aligned} & \mathbb{E}[z_{i_1;\alpha_1} z_{i_2;\alpha_2}] \\ &= \delta_{i_1 i_2} \left[g_{\alpha_1 \alpha_2} + \frac{1}{2} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1 \beta_2)(\beta_3 \beta_4)} \left(n g_{\alpha_1 \beta_1} g_{\alpha_2 \beta_2} g_{\beta_3 \beta_4} + 2 g_{\alpha_1 \beta_1} g_{\alpha_2 \beta_3} g_{\beta_2 \beta_4} \right) \right] + \mathcal{O}(v^2) \end{aligned}$$

sendo $g_{\alpha_1 \alpha_2}$ a inversa do termo quadrático $g^{\alpha_1 \alpha_2}$ de $S[z]$.

Já no correlator de 4 pontos,

$$\begin{aligned}
& \mathbb{E}[z_{i_1;\alpha_1} z_{i_2;\alpha_2} z_{i_3;\alpha_3} z_{i_4;\alpha_4}]_{\text{connected}} \\
& \equiv \mathbb{E}[z_{i_1;\alpha_1} z_{i_2;\alpha_2} z_{i_3;\alpha_3} z_{i_4;\alpha_4}] - \mathbb{E}[z_{i_1;\alpha_1} z_{i_2;\alpha_2}] \mathbb{E}[z_{i_3;\alpha_3} z_{i_4;\alpha_4}] \\
& \quad - \mathbb{E}[z_{i_1;\alpha_1} z_{i_3;\alpha_3}] \mathbb{E}[z_{i_2;\alpha_2} z_{i_4;\alpha_4}] - \mathbb{E}[z_{i_1;\alpha_1} z_{i_4;\alpha_4}] \mathbb{E}[z_{i_2;\alpha_2} z_{i_3;\alpha_3}] \\
& = \delta_{i_1 i_2} \delta_{i_3 i_4} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1 \beta_2)(\beta_3 \beta_4)} g_{\alpha_1 \beta_1} g_{\alpha_2 \beta_2} g_{\alpha_3 \beta_3} g_{\alpha_4 \beta_4} \\
& \quad + \delta_{i_1 i_3} \delta_{i_2 i_4} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1 \beta_3)(\beta_2 \beta_4)} g_{\alpha_1 \beta_1} g_{\alpha_3 \beta_3} g_{\alpha_2 \beta_2} g_{\alpha_4 \beta_4} \\
& \quad + \delta_{i_1 i_4} \delta_{i_2 i_3} \sum_{\beta_1, \dots, \beta_4 \in D} v^{(\beta_1 \beta_4)(\beta_2 \beta_3)} g_{\alpha_1 \beta_1} g_{\alpha_4 \beta_4} g_{\alpha_2 \beta_2} g_{\alpha_3 \beta_3} \\
& \quad + \mathcal{O}(v^2).
\end{aligned}$$

Comparando os slides anteriores com as contas feitas na subseção anterior, estabelecendo:

$$g^{\alpha_1 \alpha_2} = G_{(2)}^{\alpha_1 \alpha_2} + O\left(\frac{1}{n_1}\right)$$
$$v^{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)} = \frac{1}{n_1} V_{(2)}^{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)} + O\left(\frac{1}{n_1^2}\right)$$

reproduzimos os correlatores de 2ª camada até ordem $\frac{1}{n_1}$, com a distribuição marginal

$$p(z^{(2)} \mid D) = \frac{1}{Z} e^{-S[z^{(2)}]}$$

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reproduzimos os correlatores de 2^{a} camada até ordem $\frac{1}{n_1}$, com a distribuição marginal

$$p(z^{(2)} \mid D) = \frac{1}{Z} e^{-S[z^{(2)}]}$$

Obs: aqui,

$$V_{(2)}^{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)} = \sum_{\beta_1, \dots, \beta_4} G_{(2)}^{\alpha_1 \beta_1} G_{(2)}^{\alpha_2 \beta_2} G_{(2)}^{\alpha_3 \beta_3} G_{(2)}^{\alpha_4 \beta_4} V_{(\beta_1 \beta_2)(\beta_3 \beta_4)}^{(2)}$$

Introdução

4.1 - 1ª Camada: A boa e velha Gaussiana

“Wick this way”, derivação combinatória

”Hubbard-Stratonovich this way”: derivação algébrica via ação

Ação Gaussiana em ação

Seção 4.2 - 2ª Camada: A gênese da não-gaussianidade

Introdução

Distribuição condicional de segunda camada

Wick Wick Wick: derivação combinatória

Schwinger-Dyson this way: derivação algébrica

Voltamos à distribuição marginalizada:

$$p(z^{(2)} | D) = \int \left[\prod_{i;\alpha} dz_{i;\alpha}^{(1)} \right] p(z^{(1)} | z^{(2)}) p(z^{(1)} | D).$$

Voltamos à distribuição marginalizada:

$$p(z^{(2)} | D) = \int \left[\prod_{i;\alpha} dz_{i;\alpha}^{(1)} \right] p(z^{(1)} | z^{(2)}) p(z^{(1)} | D).$$

Em subseção anterior, vimos que:

$$p(z^{(2)} | z^{(1)}) = \frac{1}{|2\pi \widehat{G}^{(2)}|^{\frac{n_2}{2}}} \exp \left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2}^{n_2} \widehat{G}_{(2)}^{\alpha_1\alpha_2} z_{j;\alpha_1}^{(2)} z_{j;\alpha_2}^{(2)} \right]$$

Juntando as pecinhas:

$$\begin{aligned} p(z^{(2)} | D) &= \int \left[\prod_{i,\alpha} dz_{i;\alpha}^{(1)} \right] p(z^{(1)} | D) \frac{\exp \left(-\frac{1}{2} \sum_{j=1}^{n_2} \sum_{\alpha_1, \alpha_2 \in D} \widehat{G}_{(2)}^{\alpha_1 \alpha_2} z_{j,\alpha_1}^{(2)} z_{j,\alpha_2}^{(2)} \right)}{|2\pi \widehat{G}^{(2)}|^{\frac{n_2}{2}}} \\ &= \int \left[\prod_{i,\alpha} dz_{i;\alpha}^{(1)} \right] p(z^{(1)} | D) \frac{\exp \left(-\frac{1}{2} z^T \widehat{G}^{-1} z \right)}{|2\pi \widehat{G}|^{\frac{n_2}{2}}} \end{aligned}$$

Obs: daqui em diante, vamos omitir o sobrescrito (2) das G (aparece sempre).

Ideia: partir da decomposição

$$\widehat{G}_{\alpha_1\alpha_2} = G_{\alpha_1\alpha_2} + \widehat{\Delta G}_{\alpha_1\alpha_2},$$

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e mergulhar de cabeça ~~na loucura~~ nas expansões de Taylor.

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Primeiro: usamos a aproximação

$$\widehat{G}^{-1} = G^{-1} - G^{-1}\widehat{\Delta G}G^{-1} - G^{-1}\widehat{\Delta G}G^{-1}\widehat{\Delta G}G^{-1} + O(\Delta^3).$$

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$$\widehat{G}^{-1} = G^{-1} - G^{-1}\widehat{\Delta}GG^{-1} - G^{-1}\widehat{\Delta}GG^{-1}\widehat{\Delta}GG^{-1} + O(\Delta^3).$$

$$((M + N)^{-1} = M^{-1} - M^{-1}NM^{-1} + M^{-1}NM^{-1}NM^{-1} + O(N^3).)$$

Assim,

$$\begin{aligned} & \exp\left(-\frac{1}{2}z^T \widehat{G}^{-1} z\right) \\ &= \exp\left(-\frac{1}{2}z^T G^{-1} z\right) \\ &\times \exp\left\{1 + \frac{1}{2}z^T G^{-1} \widehat{\Delta} G^{-1} z - \frac{1}{2}z^T G^{-1} \widehat{\Delta} G^{-1} \widehat{\Delta} G^{-1} z\right\} \end{aligned}$$

Agora, usamos a expansão de Taylor em $\widehat{\Delta G}$:

$$\exp(x) \approx 1 + x + \frac{x^2}{2} + O(x^3)$$

Temos então:

$$\begin{aligned} & \exp\left(-\frac{1}{2}z^T \widehat{G}^{-1} z\right) \\ &= \exp\left(-\frac{1}{2}z^T G^{-1} z\right) \\ &\times \left\{1 + \frac{1}{2}z^T G^{-1} \widehat{\Delta G} G^{-1} z \right. \\ &\quad - \frac{1}{2}z^T G^{-1} \widehat{\Delta G} G^{-1} \widehat{\Delta G} G^{-1} \\ &\quad \left. + \frac{1}{2!} \left(\frac{1}{2}\right)^2 (z^T G^{-1} \widehat{\Delta G} G^{-1} z)^2 + O(\Delta^3)\right\} \end{aligned}$$

Usando na expressão do determinante:

$$\begin{aligned}
 |2\pi\widehat{G}|^{\frac{n_2}{2}} &= \int \left[\prod_{i;\alpha} dz_{i;\alpha} \right] \exp\left(-\frac{1}{2}z^T G z\right) \\
 &= \int \left[\prod_{i;\alpha} dz_{i;\alpha} \right] \exp\left(-\frac{1}{2}z^T G^{-1} z\right) \times \left\{ 1 + \frac{1}{2}z^T G^{-1} \widehat{\Delta G} G^{-1} z \right. \\
 &\quad \left. - \frac{1}{2}z^T G^{-1} \widehat{\Delta G} G^{-1} \widehat{\Delta G} G^{-1} + \frac{1}{2!} \left(\frac{1}{2}\right)^2 (z^T G^{-1} \widehat{\Delta G} G^{-1} z)^2 + O(\Delta^3) \right\} \\
 &= \sqrt{|(2\pi G)|^{n_2}} \left[1 + \frac{n_2}{2} \sum_{\beta_1, \beta_2 \in D} \widehat{\Delta G}_{\beta_1 \beta_2} G^{\beta_1 \beta_2} \right. \\
 &\quad \left. + \sum_{\beta_1, \dots, \beta_4 \in D} \widehat{\Delta G}_{\beta_1 \beta_2} \widehat{\Delta G}_{\beta_3 \beta_4} \left(\frac{n_2^2}{8} G^{\beta_1 \beta_2} G^{\beta_3 \beta_4} - \frac{n_2}{4} G^{\beta_1 \beta_3} G^{\beta_2 \beta_4} \right) + O(\Delta^3) \right]
 \end{aligned}$$

$$\begin{aligned}
p(z^{(2)} \mid D) &= \frac{1}{|2\pi G^{(2)}|^{\frac{n_2}{2}}} \exp\left(-\frac{1}{2} \sum_{j=1}^{n_2} \sum_{\alpha_1, \alpha_2 \in D} G_{(2)}^{\alpha_1 \alpha_2} z_{j; \alpha_1}^{(2)} z_{j; \alpha_2}^{(2)}\right) \\
&\times \left\{ \left[1 + \mathcal{O}\left(\frac{1}{n_1}\right)\right] + n_2 \sum_{i=1}^{n_2} \sum_{\alpha_1, \alpha_2 \in D} \left[\mathcal{O}\left(\frac{1}{n_1}\right)\right] z_{i; \alpha_1}^{(2)} z_{i; \alpha_2}^{(2)} \right. \\
&\quad \left. + \frac{1}{8n_1} n_2 \sum_{i_1, i_2=1}^{n_2} \sum_{\alpha_1, \dots, \alpha_4 \in D} V_{(2)}^{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)} z_{i_1; \alpha_1}^{(2)} z_{i_1; \alpha_2}^{(2)} z_{i_2; \alpha_3}^{(2)} z_{i_2; \alpha_4}^{(2)} \right\} \\
&+ \mathcal{O}\left(\frac{1}{n_1^2}\right)
\end{aligned}$$

Obrigado!

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Repositório com os materiais da apresentação:
<https://github.com/felipekriffel/Mestrado>